RAFAEL MORETTI-KLEIN

Senior Backend Engineer | Python, Go

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PROFESSIONAL SUMMARY

Senior backend engineer with 21 years of experience building production services and large-scale distributed systems in Python, Go and C++. Designed and wrote most of a Go entitlement service handling 120k requests per second, and led technical work on a playback API and a migration from a Python 2 monolith to around 60 Python and Go services; playback API availability improved from 99.5% to 99.95%. Experienced with PostgreSQL, backend APIs, Kafka, Kubernetes, Terraform and observability tooling. Fluent in English and experienced in documenting architecture.

PROFESSIONAL EXPERIENCE

Director of Engineering, Platform | Streamhaus GmbH, Munich *2019 – August 2026*

- Led the technical migration of a Python 2 monolith into approximately 60 Python and Go services deployed on GKE; playback API availability improved from 99.5% to 99.95%.
- Established a service template using Python/FastAPI or Go, OpenTelemetry tracing, Prometheus metrics and Terraform modules.
- Built an internal cost dashboard using Python, BigQuery billing exports and Looker Studio; rightsizing and committed-use discounts reduced cloud spend by 31% (€4.2M per year).
- Led the platform organisation responsible for identity, playback API, billing integration and developer platform.
- Introduced an RFC process, architecture review board and on-call guidelines adopted company-wide.

Principal Engineer | Streamhaus GmbH *2015 – 2019*

- Led technical work on the playback API using Python, Tornado and an internal framework similar to FastAPI, with Cassandra and Redis.
- Designed and wrote most of a Go entitlement service for subscriptions, regions and devices using gRPC, Cassandra, Redis and Kubernetes; it handled peak traffic of 120k requests per second with p99 latency under 15 ms.
- Co-authored the company's service template and Python style guide.

Senior Software Engineer | Nordfels Telematik AG, Stuttgart *2010 – 2015*

- Built backend systems for fleet telematics, ingesting GPS data from 300k vehicles using C++, Python, PostgreSQL/PostGIS and RabbitMQ.
- Led a team of six developers on the route-history product.
- Co-inventor of a granted patent for adaptive GPS sampling in vehicle fleets.

Software Engineer | Dataquelle Systems GmbH, Karlsruhe *2005 – 2010*

- Developed C++ and Python software for industrial data acquisition on Linux and embedded x86.
- Built a Python test harness used by QA for eight years.

SELECTED PROJECTS & OPEN SOURCE

- Open-source maintainer, Python retry library (2012–2018): Maintained a library with approximately 1.5k GitHub stars; used pytest and tox.
- Entitlement Service (Streamhaus, 2016–2017): Go, gRPC, Cassandra, Redis and Kubernetes; 120k requests per second at peak.
- Monolith to Services Migration (Streamhaus, 2019–2023): Python, FastAPI, Go, GKE, Terraform, OpenTelemetry and Prometheus.

EDUCATION

Dipl.-Inf. (equivalent to MSc), Computer Science | Rhine Valley University, 2005

TECHNICAL SKILLS

Languages: Python, Go, C++
Backend and APIs: FastAPI, Django, Flask, Tornado, asyncio, gRPC, Protobuf
Databases and messaging: PostgreSQL, PostGIS, Cassandra, Redis, Kafka, RabbitMQ
Cloud and infrastructure: GCP (GKE, Pub/Sub, BigQuery), Kubernetes, Terraform; AWS (EC2, S3)
Testing and observability: pytest, tox, Prometheus, Grafana, OpenTelemetry
Other: Distributed systems design, architecture reviews, technical hiring, people management

LANGUAGES

German (native), Italian (native), English (fluent)

PUBLICATIONS & TALKS

- "Running Python at 100k rps" — conference talk, 2018
- "Cost is a feature" — internal platform engineering summit keynote, 2023